



Applied Stats II

Replication Project

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The Paper

“Descriptive and Substantive Representation in Congress: Evidence from 80,000 Congressional Inquiries” by Kenneth Lowande, Melinda Ritchie, and Erinn Lauterbach (2019).

“Do representatives follow up on the policy interests of women, minorities, and veterans long after a bill's passage? Do they monitor agencies to ensure the rights of these communities are protected?” (p. 645)

- Authors submitted Freedom of Information Act (FOIA) requests to federal agencies, receiving approximately 88,000 records of contact made by legislators to bureaucrats.
- Their aim was to see which legislators contacted a federal agency on behalf of veterans, women, or racial/ethnic minorities at least once per congressional session.
- They hand-coded communications (1 for contact on behalf of a particular protected group, 0 otherwise) according to strict criteria for inclusion, such that they are more likely to *undercount* representative measures by congress members.
- Drawing on information from CQ Press' congressional member profiles, they coded identities of lawmakers according to their belonging to legally-protected groups and aggregated the data to congressperson-congressional session observations.

Data Layout

Outline of the data structure:

Index	Congressman ID	Congress	r.vet	r.fem	r.race	District Ideology
1	092	111th	0	0	1	Liberal
2	092	112th	0	0	1	Liberal
3	093	110th	0	1	0	Moderate
4	093	111th	0	1	1	Moderate

Table 3: Military Service and Veterans Representation

	Dependent Variable:			
	All Contact		110th - 112th Cong.	
	(1)	(2)	(3)	(4)
Veteran (Any)	0.029 (0.024)		0.047 (0.028)	
Veteran (Excluding Reservists)		0.068 (0.030)		0.072 (0.032)
Commonspace Ideology	-0.016 (0.022)	-0.016 (0.022)	-0.030 (0.025)	-0.030 (0.025)
ln(Veteran Expenditures)	0.073 (0.015)	0.073 (0.015)	0.070 (0.019)	0.071 (0.019)
Congress Fixed Effects	✓	✓	✓	✓
N	2,194	2,194	1,654	1,654
Adjusted R^2	0.15	0.16	0.09	0.09

Table 6: Race/Ethnicity and Minority Representation

	Dependent Variable:			
	All Contact		110th - 112th Cong.	
	(1)	(2)	(3)	(4)
Nonwhite	0.091*** (0.034)	0.060 (0.039)	0.090** (0.037)	0.078* (0.043)
District Ideology	-0.153*** (0.041)		-0.203*** (0.043)	
District % White		-0.276*** (0.075)		-0.264*** (0.082)
Congress Fixed Effects	✓	✓	✓	✓
N	2,194	2,194	1,654	1,654
Adjusted R^2	0.11	0.12	0.11	0.11

Table 5: Gender and Women's Representation

	(1)	(2)
Female	0.079 (0.020)	0.083 (0.021)
Commonspace Ideology	-0.196 (0.015)	
District Ideology		-0.256 (0.024)
Congress Fixed Effects	✓	✓
N	2,194	2,194
Adjusted R^2	0.21	0.19

*All coefficients are significant to at least the $p < 0.05$ level.

Findings

Running OLS models on representation tendencies of legislators with pertinent controls, the authors found significant relationships between legislator identity/background and the probability that they reach out to a federal agency on the behalf of protected groups with the same identity/background.

In other terms, veterans (especially non-reservists) were more likely to advocate for veterans, women were more likely to advocate for women, and racial/ethnic minorities were more likely to advocate for racial/ethnic minorities.

This supports the growing literature outlining the link between descriptive and substantive representation.

My Contributions



Gender / Race Interaction

Do female Congress members advocate for racial minorities more than their male counterparts?

(OLS models)



Logistic Modeling

What impact does each variable of interest have on the odds of a representative advocating for a particular protected group?

(Logit models)



Visualizing Substantive Representation

How does “substantive” representation differ geographically & over time?

(Proportions of Legislators)

Women Advocating for Racial/Ethnic Minorities?

Table 7: Gender and Minority Representation

	All Contact		110th - 112th Cong.	
	(1)	(2)	(3)	(4)
Female	-0.004 (0.028)	0.007 (0.028)	-0.003 (0.031)	0.015 (0.031)
District Ideology	-0.201*** (0.041)		-0.250*** (0.042)	
District % White		-0.346*** (0.062)		-0.353*** (0.068)
Congress Fixed Effects	✓	✓	✓	✓
N	2194	2194	1654	1654
Adjusted R^2	0.1	0.1	0.11	0.11

Note:

* $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$

There is no evidence of a significant relationship between legislator gender and odds of advocating for racial minorities.

Racial/Ethnic Minorities Advocating for Women?

... Likewise, there is no evidence of a significant relationship between legislator race (White vs. Nonwhite) and their odds of advocating for women.

Table 8: Race/Ethnicity and Women's Representation

	(1)	(2)
Nonwhite	0.020 (0.022)	0.028 (0.024)
Commonspace Ideology	-0.204*** (0.016)	
District Ideology		-0.267*** (0.025)
Congress Fixed Effects	✓	✓
N	2194	2194
Adjusted R^2	0.2	0.18

Note:

* $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$

Logit Models: Formulas

Veterans

Representation dummy regressed on veteran status of legislator, Congress chamber and session, Commonsense Ideology score, and log of district expenditures on veterans.

```
glm(r.vet ~ vet.mainbranch+chamber+dwnom1+as.factor(cong)+log(vets.exp),  
     data = main, family = binomial(link = "logit"))
```

Women

Representation dummy regressed on gender of legislator, Congress chamber and session, Commonsense Ideology score, and district mean ideology.

```
glm(r.fem ~ female+chamber+dwnom1+as.factor(cong)+tw.ideo.mean,  
     data = main, family = binomial(link = "logit"))
```

Ethnic/Racial Minorities

Representation dummy regressed on race of legislator, Congress chamber and session, Commonsense Ideology score, district mean ideology, and percent of population White.

```
glm(r.race ~ nonwhite+chamber+as.factor(cong)+dwnom1+tw.ideo.mean+white.pop,  
     data = main, family = binomial(link = "logit"))
```

*Code here is heavily edited for presentation readability, two logit models run for veterans (one excluding reservists)

Logit Results Support Original Study Findings For Veterans' Representation

Veterans

Original Diff: **+0.06** | **+0.09** Logit OR: **1.87** | **1.60**

Non-vet odds (0.25/0.75) * **1.87** | **1.60** = Vet odds = 0.62 | 0.53

Vets probability = (0.62/1.62) | (0.53/1.53) = 0.38 | 0.34

Logit OR applied Diff: **+0.13** | **+0.09**

Difference with Logit OR applied matches model for Non-Reservists only, but significant positive sign for “Any” category constitutes support for findings.

Table 2: Representation Differences across Members with Military Service

Any Veteran	Nonveterans	Veterans	Difference (95 % CI)
All Legislators (n = 2194)	0.25	0.31	+0.06 [0.1, 0.01]
Excluding Reservists	Nonveterans	Veterans	Difference (95 % CI)
All Legislators (n = 1194)	0.25	0.34	+0.09 [0.15, 0.04]

Odds Ratio Coefficients for Logit Model of Veteran Representation

	(1)	(2)
Constant	0.003 *** (0.004)	0.003*** (0.01)
111th Congress	0.70* (0.14)	0.70* (0.14)
112th Congress	0.73 (0.15)	0.72*** (0.15)
Veteran (Any)	1.87*** (0.36)	
Veteran (Excluding Reservists)		1.60** (0.34)
ln(Veteran Expenditures)	1.32** (0.17)	1.31** (0.17)
Senate	2.07** (0.59)	2.12*** (0.62)
Commonspace Ideology	1.24 (0.25)	1.28 (0.26)
Observations	1,654	1,654
Log Likelihood	-507.46	-510.72

Note: *p<0.1; **p<0.05; ***p<0.01

Reference categories are 110th Congress and the House of Representatives Chamber. Model excludes legislator contact regarding casework. Significance calculated based on non-transformed coefficients and their standard errors. Coefficients that lost significance through logit modeling in bold text.

*This comparison is true for the All Legislators section of their data rather than the Split-Senate delegations, as my logit model is based on All Legislators.

Logit Results Bolster Original Study Findings For Women's Representation

Women

Original Diff: **+0.12** Logit OR: **2.4**

Male odds (0.11/0.89) * **2.4** = Female odds = 0.298

Female probability = 0.298/1.298 = 0.23

Difference with Logit OR applied matches model.

Table 4: Representation Differences across Gender

	Male	Female	Difference (95 % CI)
All Legislators (n = 2194)	0.11	0.23	+0.12 [0.07, 0.16]

Odds Ratio Coefficients for Logit Model of Women's Representation

Constant	0.02*** (0.01)
110th Congress	0.36*** (0.14)
111th Congress	0.50*** (0.17)
112th Congress	0.49*** (0.18)
Female	2.40*** (0.74)
District Ideology	1.32*** (0.82)
Senate	5.20*** (1.51)
Commonspace Ideology	0.25*** (0.12)
Observations	2,194
Log Likelihood	-248.55

Note: *p<0.1; **p<0.05; ***p<0.01

Reference categories are 109th Congress and the House of Representatives Chamber. Model excludes legislator contact regarding casework.

*This comparison is true for the All Legislators section of their data rather than the Split-Senate delegations, as my logit model is based on All Legislators.

Logit Results Don't Support Original Study Findings For Racial/Ethnic Minority Representation

Racial/Ethnic Minorities

Original Diff: **+0.09**

Logit OR: **1.4**

Not statistically significant!

Loss of significance for Nonwhite identity of legislator as predictor for advocacy on behalf of racial/ethnic minorities.

Could be artifact of data differences: I drew strictly from the non-casework observations of the complete-case data subset.

Odds Ratio Coefficients for Logit Model of Minority Representation

Constant	0.36** (0.15)
Nonwhite	1.37 (0.30)
District Ideology	0.39*** (0.14)
District % White	0.50 (0.25)
111th Congress	0.99 (0.13)
112th Congress	1.11 (0.16)
Senate	5.65*** (0.86)
Commonspace Ideology	0.96 (0.21)
Observations	1,654
Log Likelihood	-868.25

Note: *p<0.1; **p<0.05; ***p<0.01

Reference categories are 110th Congress and the House of Representatives Chamber. Model excludes legislator contact regarding casework. Significance calculated based on non-transformed coefficients and their standard errors. Coefficients that lost significance through logit modeling in bold text.

Logit Models: Example Comparisons

Holding all else constant, on average...



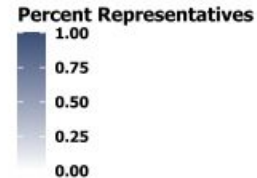
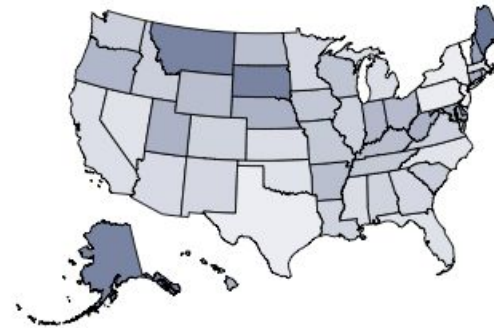
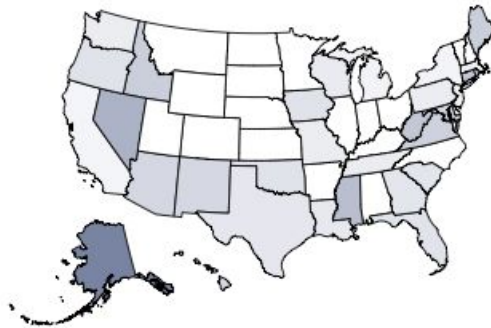
- Veteran legislators at this time have 60% higher odds of intervening than non-veteran legislators during these congresses.
- Female legislators at this time had more than double the odds of intervening on behalf of women than male legislators.
- Districts with 0.1 points more conservative mean ideologies had representatives with 3% greater odds of intervening.
- Non-significance of coefficients for selected data deters us from using this in any comparisons.

Veterans' Representation

110th Congress

111th Congress

112th Congress



Plots indicate proportion of legislators contacting federal agencies on behalf of veterans (excluding casework) at least once per congressional term for the 110th, 111th, and 112th Congresses. Darker shades of blue indicate higher values.

Women's Representation

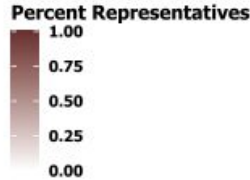
110th Congress



111th Congress



112th Congress



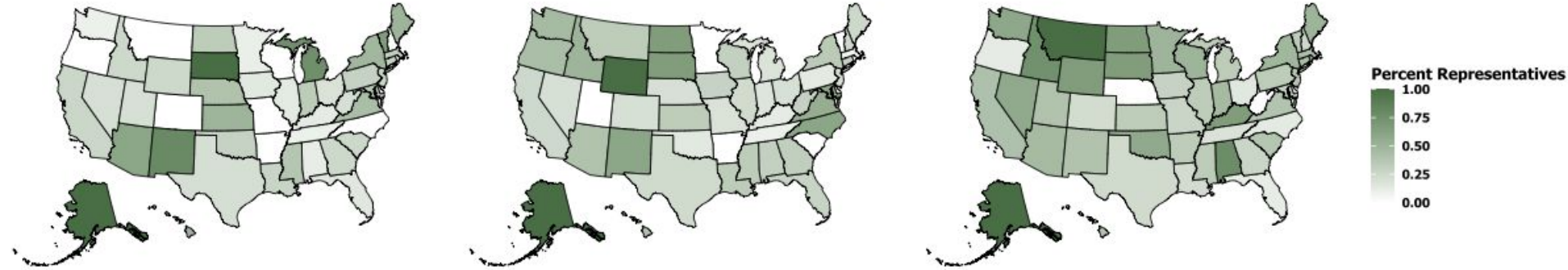
Plots indicate proportion of legislators contacting federal agencies on behalf of women (excluding casework) at least once per congressional term for the 110th, 111th, and 112th Congresses. Darker shades of red indicate higher values.

Racial/Ethnic Minorities' Representation

110th Congress

111th Congress

112th Congress



Plots indicate proportion of legislators contacting federal agencies on behalf of racial/ethnic minorities (excluding casework) at least once per congressional term for the 110th, 111th, and 112th Congresses. Darker shades of green indicate higher values.

Thanks

Do you have any questions?

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Sources Utilized in Replication Project

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